Eigen Values and Eigen Vectors

Linear Algebra - 24DS

AH Sheikh

QUEST - 24DS

Definition and Key Idea

Definition

For a square matrix A, a **non-zero** vector $\mathbf{v}$ is an **eigenvector** if:

$$A\mathbf{v} = \lambda\mathbf{v}$$

where λ (a scalar) is the corresponding **eigenvalue**.

Definition and Key Idea

Definition

For a square matrix A, a **non-zero** vector $\mathbf{v}$ is an **eigenvector** if:

$$A\mathbf{v} = \lambda\mathbf{v}$$

where λ (a scalar) is the corresponding **eigenvalue**.

Geometric meaning: The linear transformation only scales $\mathbf{v}$; direction unchanged (or reversed if $\lambda < 0$).

Definition and Key Idea

Definition

For a square matrix A, a **non-zero** vector $\mathbf{v}$ is an **eigenvector** if:

$$A\mathbf{v} = \lambda\mathbf{v}$$

where λ (a scalar) is the corresponding **eigenvalue**.

Geometric meaning: The linear transformation only scales $\mathbf{v}$; direction unchanged (or reversed if $\lambda < 0$).

How to find eigenvalues

Solve the **characteristic equation**:

$$\det(A - \lambda I) = 0$$

The polynomial in λ has degree n (size of matrix).

Definition and Key Idea

Definition

For a square matrix A, a **non-zero** vector $\mathbf{v}$ is an **eigenvector** if:

$$A\mathbf{v} = \lambda\mathbf{v}$$

where λ (a scalar) is the corresponding **eigenvalue**.

Geometric meaning: The linear transformation only scales $\mathbf{v}$; direction unchanged (or reversed if $\lambda < 0$).

How to find eigenvalues

Solve the **characteristic equation**:

$$\det(A - \lambda I) = 0$$

The polynomial in λ has degree n (size of matrix).

2×2 Example — Fully Solved

Example

Find eigenvalues & eigenvectors of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

$2{\times}2$ Example — Fully Solved

Example

Find eigenvalues & eigenvectors of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

Step 1: $A - \lambda I = \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}$

$$\det = (4-\lambda)(3-\lambda) - (1)(2) = \lambda^2 - 7\lambda + 10 = 0$$

2×2 Example — Fully Solved

Example

Find eigenvalues & eigenvectors of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

Step 1: $A - \lambda I = \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}$

$$\det = (4-\lambda)(3-\lambda) - (1)(2) = \lambda^2 - 7\lambda + 10 = 0$$

$$(\lambda - 2)(\lambda - 5) = 0 \quad \Rightarrow \quad \lambda_1 = 2,\ \lambda_2 = 5$$

2×2 Example — Fully Solved

Example

Find eigenvalues & eigenvectors of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

Step 1: $A - \lambda I = \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}$

$$\det = (4-\lambda)(3-\lambda) - (1)(2) = \lambda^2 - 7\lambda + 10 = 0$$

$$(\lambda - 2)(\lambda - 5) = 0 \quad \Rightarrow \quad \lambda_1 = 2,\ \lambda_2 = 5$$

Step 2: Eigenvector for $\lambda = 2$:

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \ \Rightarrow \ 2x + y = 0$$

Choose $x = 1,\ y = -2$: $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

2×2 Example — Fully Solved

Example

Find eigenvalues & eigenvectors of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

Step 1: $A - \lambda I = \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}$

$$\det = (4-\lambda)(3-\lambda) - (1)(2) = \lambda^2 - 7\lambda + 10 = 0$$

$$(\lambda - 2)(\lambda - 5) = 0 \quad \Rightarrow \quad \lambda_1 = 2,\ \lambda_2 = 5$$

Step 2: Eigenvector for $\lambda = 2$:

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \ \Rightarrow \ 2x + y = 0$$

Choose $x = 1,\ y = -2$: $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

For $\lambda = 5$:

3×3 Example — Fully Solved (Part 1)

Example

$$A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

3×3 Example — Fully Solved (Part 1)

Example

$$A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

Char. eq: $\det(A - \lambda I) = 0$:

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

3×3 Example — Fully Solved (Part 1)

Example

$$A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

Char. eq: $\det(A - \lambda I) = 0$:

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

Compute determinant (expand row 1):

$$(6-\lambda)\left[(3-\lambda)^2 - 1\right] + 2\left[-2(3-\lambda) + 2\right] + 2\left[2 - 2(3-\lambda)\right]$$

3×3 Example — Fully Solved (Part 1)

Example

$$A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

Char. eq: $\det(A - \lambda I) = 0$:

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

Compute determinant (expand row 1):

$$(6-\lambda)\left[(3-\lambda)^2 - 1\right] + 2\left[-2(3-\lambda)+2\right] + 2\left[2 - 2(3-\lambda)\right]$$

Simplify: $(6-\lambda)(\lambda^2 - 6\lambda + 8) + 2(-6 + 2\lambda + 2) + 2(2 - 6 + 2\lambda)$

$$= (6-\lambda)(\lambda-2)(\lambda-4) + 2(2\lambda-4) + 2(2\lambda-4)$$

3×3 Example — Fully Solved (Part 1)

Example

$$A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

Char. eq: $\det(A - \lambda I) = 0$:

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

Compute determinant (expand row 1):

$$(6-\lambda)\left[(3-\lambda)^2 - 1\right] + 2\left[-2(3-\lambda) + 2\right] + 2\left[2 - 2(3-\lambda)\right]$$

Simplify: $(6-\lambda)(\lambda^2 - 6\lambda + 8) + 2(-6 + 2\lambda + 2) + 2(2 - 6 + 2\lambda)$

$$= (6-\lambda)(\lambda-2)(\lambda-4) + 2(2\lambda-4) + 2(2\lambda-4)$$

Note $2(2\lambda - 4) + 2(2\lambda - 4) = 8(\lambda - 2)$. Factor $(\lambda - 2)$:

3×3 Example — Eigenvectors

Eigenvalues: $\lambda_1 = 2$ (mult. 2), $\lambda_2 = 8$.

For $\lambda = 2$: $A - 2I = \begin{pmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix}$

$$\text{Row reduce: } \begin{pmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Equation: $2x - y + z = 0 \quad y = 2x + z$.

3×3 Example — Eigenvectors

Eigenvalues: $\lambda_1 = 2$ (mult. 2), $\lambda_2 = 8$.

For $\lambda = 2$**:** $A - 2I = \begin{pmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix}$

$$\text{Row reduce: } \begin{pmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Equation: $2x - y + z = 0$ $y = 2x + z$. Free vars: x, z. Two independent eigenvectors:

$$x = 1, z = 0 \Rightarrow \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \qquad x = 0, z = 1 \Rightarrow \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

3×3 Example — Eigenvectors

Eigenvalues: $\lambda_1 = 2$ (mult. 2), $\lambda_2 = 8$.

For $\lambda = 2$**:** $A - 2I = \begin{pmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix}$

$$\text{Row reduce: } \begin{pmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Equation: $2x - y + z = 0 \quad y = 2x + z$. Free vars: x, z. Two independent eigenvectors:

$$x = 1, z = 0 \Rightarrow \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \qquad x = 0, z = 1 \Rightarrow \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

For $\lambda = 8$**:** $A - 8I = \begin{pmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

CS Application 1: Principal Component Analysis (PCA)

Problem: Reduce 2D data $\{(1,2),(2,3),(3,4),(4,5)\}$ to 1D while preserving max variance.

CS Application 1: Principal Component Analysis (PCA)

Problem: Reduce 2D data $\{(1,2),(2,3),(3,4),(4,5)\}$ to 1D while preserving max variance.

Step 1: Center data (mean subtract). Mean = (2.5, 3.5). Centered:

$$(-1.5,-1.5),\ (-0.5,-0.5),\ (0.5,0.5),\ (1.5,1.5)$$

CS Application 1: Principal Component Analysis (PCA)

Problem: Reduce 2D data $\{(1,2),(2,3),(3,4),(4,5)\}$ to 1D while preserving max variance.
Step 1: Center data (mean subtract). Mean = (2.5, 3.5). Centered:

$$(-1.5,-1.5),\ (-0.5,-0.5),\ (0.5,0.5),\ (1.5,1.5)$$

Step 2: Covariance matrix $C = \frac{1}{n-1}X^TX$ (using centered data matrix X):

$$X = \begin{pmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{pmatrix}$$

$$C = \frac{1}{3}\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix} = \begin{pmatrix} 5/3 & 5/3 \\ 5/3 & 5/3 \end{pmatrix}$$

CS Application 1: Principal Component Analysis (PCA)

Problem: Reduce 2D data $\{(1,2),(2,3),(3,4),(4,5)\}$ to 1D while preserving max variance.

Step 1: Center data (mean subtract). Mean = (2.5, 3.5). Centered:

$$(-1.5,-1.5),\ (-0.5,-0.5),\ (0.5,0.5),\ (1.5,1.5)$$

Step 2: Covariance matrix $C = \frac{1}{n-1}X^TX$ (using centered data matrix X):

$$X = \begin{pmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{pmatrix}$$

$$C = \frac{1}{3}\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix} = \begin{pmatrix} 5/3 & 5/3 \\ 5/3 & 5/3 \end{pmatrix}$$

Step 3: Eigenvalues of C: $\det\begin{pmatrix} 5/3-\lambda & 5/3 \\ 5/3 & 5/3-\lambda \end{pmatrix} = 0$

$$(5/3-\lambda)^2 - (5/3)^2 = 0 \Rightarrow \lambda(\lambda - 10/3) = 0$$

CS Application 1: Principal Component Analysis (PCA)

Problem: Reduce 2D data $\{(1,2),(2,3),(3,4),(4,5)\}$ to 1D while preserving max variance.

Step 1: Center data (mean subtract). Mean = (2.5, 3.5). Centered:

$$(-1.5,-1.5),\ (-0.5,-0.5),\ (0.5,0.5),\ (1.5,1.5)$$

Step 2: Covariance matrix $C = \frac{1}{n-1}X^TX$ (using centered data matrix X):

$$X = \begin{pmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{pmatrix}$$

$$C = \frac{1}{3}\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix} = \begin{pmatrix} 5/3 & 5/3 \\ 5/3 & 5/3 \end{pmatrix}$$

Step 3: Eigenvalues of C: $\det\begin{pmatrix} 5/3-\lambda & 5/3 \\ 5/3 & 5/3-\lambda \end{pmatrix} = 0$

$$(5/3-\lambda)^2 - (5/3)^2 = 0 \Rightarrow \lambda(\lambda - 10/3) = 0$$

CS Application 2: PageRank Algorithm

Problem: Rank 3 web pages with links:

$$1 \to 2,\ 2 \to 3,\ 3 \to 1,\ 3 \to 2$$

CS Application 2: PageRank Algorithm

Problem: Rank 3 web pages with links:

$$1 \to 2,\ 2 \to 3,\ 3 \to 1,\ 3 \to 2$$

Step 1: Adjacency matrix G (rows = from, cols = to):

$$G = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

CS Application 2: PageRank Algorithm

Problem: Rank 3 web pages with links:

$$1 \to 2,\ 2 \to 3,\ 3 \to 1,\ 3 \to 2$$

Step 1: Adjacency matrix G (rows = from, cols = to):

$$G = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Step 2: Stochastic matrix M (divide by out-degree): Page 1 outdeg=1, page2 outdeg=1, page3 outdeg=2.

$$M = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

CS Application 2: PageRank Algorithm

Problem: Rank 3 web pages with links:

$$1 \to 2,\ 2 \to 3,\ 3 \to 1,\ 3 \to 2$$

Step 1: Adjacency matrix G (rows = from, cols = to):

$$G = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Step 2: Stochastic matrix M (divide by out-degree): Page 1 outdeg=1, page2 outdeg=1, page3 outdeg=2.

$$M = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

Step 3: PageRank = eigenvector for $\lambda = 1$ of M^T (or left eigenvector of M): Solve $(M^T - I)\mathbf{r} = 0$: $\begin{pmatrix} -1 & 0 & 1/2 \\ 1 & -1 & 1/2 \\ 0 & 1 & -1 \end{pmatrix} \mathbf{r} = 0$

CS Application 2: PageRank Algorithm

Problem: Rank 3 web pages with links:

$$1 \to 2,\ 2 \to 3,\ 3 \to 1,\ 3 \to 2$$

Step 1: Adjacency matrix G (rows = from, cols = to):

$$G = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Step 2: Stochastic matrix M (divide by out-degree): Page 1 outdeg=1, page2 outdeg=1, page3 outdeg=2.

$$M = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

Step 3: PageRank = eigenvector for $\lambda = 1$ of M^T (or left eigenvector of M): Solve $(M^T - I)\mathbf{r} = 0$: $\begin{pmatrix} -1 & 0 & 1/2 \\ 1 & -1 & 1/2 \\ 0 & 1 & -1 \end{pmatrix} \mathbf{r} = 0$ From row1:

Summary & Key Takeaways

Computational process:

1. $\det(A - \lambda I) = 0 \rightarrow$ eigenvalues
2. $(A - \lambda I)\mathbf{v} = 0 \rightarrow$ eigenvectors

Properties:

- Trace = sum of eigenvalues
- Determinant = product

CS Applications:

- PCA (dim reduction)
- PageRank (web search)
- Spectral clustering
- Face recognition (Eigenfaces)
- Google's matrix computations

Summary & Key Takeaways

Computational process:

1. $\det(A - \lambda I) = 0 \rightarrow$ eigenvalues
2. $(A - \lambda I)\mathbf{v} = 0 \rightarrow$ eigenvectors

Properties:

- Trace = sum of eigenvalues
- Determinant = product

CS Applications:

- PCA (dim reduction)
- PageRank (web search)
- Spectral clustering
- Face recognition (Eigenfaces)
- Google's matrix computations

Always verify

$A\mathbf{v} = \lambda\mathbf{v}$ for your solutions!

Practice Exercise

Find eigenvalues/vectors of this 2×2 matrix used in Markov chains:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix}$$

Practice Exercise

Find eigenvalues/vectors of this 2×2 matrix used in Markov chains:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix}$$

Solution outline:

$$\det \begin{pmatrix} 0.7-\lambda & 0.3 \\ 0.2 & 0.8-\lambda \end{pmatrix} = (0.7-\lambda)(0.8-\lambda) - 0.06 = \lambda^2 - 1.5\lambda + 0.5 = 0$$

$$\lambda = \frac{1.5 \pm \sqrt{2.25-2}}{2} = \frac{1.5 \pm 0.5}{2} \Rightarrow \lambda_1 = 1,\ \lambda_2 = 0.5$$

Practice Exercise

Find eigenvalues/vectors of this 2×2 matrix used in Markov chains:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix}$$

Solution outline:

$$\det \begin{pmatrix} 0.7 - \lambda & 0.3 \\ 0.2 & 0.8 - \lambda \end{pmatrix} = (0.7-\lambda)(0.8-\lambda) - 0.06 = \lambda^2 - 1.5\lambda + 0.5 = 0$$

$$\lambda = \frac{1.5 \pm \sqrt{2.25 - 2}}{2} = \frac{1.5 \pm 0.5}{2} \Rightarrow \lambda_1 = 1,\ \lambda_2 = 0.5$$

Eigenvector for $\lambda = 1$: $(0.6, 0.4)$ — stationary distribution.